

Lucas Lemos

Software Engineer

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I like to find the most optimal and effective ways for machines to solve our problems.

Professional Summary

Software Engineer with 4+ years of experience specializing in Rust and distributed systems. Proven ability to design and implement low-latency, high-throughput services on AWS, optimize cloud infrastructure, and develop resilient data pipelines. Expertise spans Cryptography, Backend, Distributed Systems, Blockchain, CI/CD automation, and successful mentorship of junior developers, delivering robust and scalable solutions.

EXPERIENCE

Senior Software Engineer, NDA Contract, Remote

Jan 2026 to May 2026

- Worked on high-performance Rust APIs for automated digital asset trading using Solana.
- Implemented secure OAuth authentication and payment gateway integrations.
- Implemented real-time digital asset monitoring.
- Built a caching layer using Redis. Reducing system load on peak times by 40%.
- Set up an observability dashboard tracking performance and load using Prometheus and Grafana.
- Minimized latency in bots from ms to μ s by optimizing concurrency and minimizing memory allocation.
- Collaborated with engineering teams to deliver scalable trading functionality and establish Rust development best practices.
- Written documentation for features, services, and best practices.
- Provided mentorship for Junior developers.

Software Engineer, Deckers Brands, Goleta

Aug 2024 to Dec 2025

- Architected an event-driven, multi-region data processing platform in Rust using AWS Lambda, API Gateway, SQS, DynamoDB Streams, and S3, processing over 1M events/day with automated dead-letter recovery ensuring zero data loss, enabling reliable near real-time data synchronization across four global regions.
- Built a high-throughput ingestion and data access layer using Rust, DynamoDB, PostgreSQL/Aurora, and SeaORM, handling 2,000 requests/second with atomic real-time updates, reducing data propagation latency and improving consistency across distributed services.
- Designed and implemented an end-to-end secure file delivery pipeline using AWS Lambda, S3, Secrets Manager, AWS Transfer Family (SFTP), and PGP encryption, delivering authenticated partner file transfers in under 30 seconds, replacing manual integration workflows with a secure, automated delivery process.
- Developed Rust integrations with multiple external third-party APIs, using async HTTP clients (Tokio, Request), OAuth-style token authentication with automatic refresh via AWS Secrets Manager, JSON serialization, and retry/backoff handling for transient failures, replacing daily batch file processing with real-time third-party integrations.

Lead Developer, PrivID, Toronto

Jan 2023 to Jul 2024

- Built high-performance APIs in Rust, focusing on low-latency and scalability for critical services.
- Implemented Zero-Knowledge systems (ZK-SNARKs / ZK-STARKs) for secure and verifiable computation, enhancing data privacy and integrity.
- Developed Rust-based WebAssembly (WASM) modules for seamless frontend integration and efficient client-side processing.
- Deployed containerized services with Docker and Kubernetes in AWS environments, optimizing resource utilization and scalability.
- Designed CI/CD pipelines in GitLab to automate testing and deployments, reducing release cycles by 25%.
- Mentored junior developers and contributed to a patent-backed system design, fostering team growth and innovation.

Blockchain Developer, NEAR Foundation, Remote

Jan 2022 to Aug 2022

- Assisted companies in launching NFTs and other services in Web-3, navigating new technological landscapes.
- Delivered training courses and workshops about Rust development with an emphasis on blockchain, upskilling fellow developers.
- Created and delivered interactive and engaging training materials, including lesson plans, presentations, and hands-on exercises.
- Offered feedback and contributions to the SDK of NEAR protocol, enhancing platform functionality and developer experience.
- Wrote tutorials and articles on various subjects related to blockchain, including smart contract development and deployment.
- Implemented listeners for tracking blockchain events in real time, ensuring data consistency and responsiveness.

Web Developer Freelancer (Remote), IT Services and IT Consulting, Divinópolis

Jan 2020 to Jul 2022

- Implemented features for React Full-stack applications using NodeJs and Rust for various clients.
- Deployed Rust services in Google Cloud and AWS environments.
- Implemented WebAssembly dynamic libraries for the browser to process heavy data workloads efficiently.
- Fixed critical Javascript bugs in client-side applications, improving user experience.
- Created simple web interfaces using HTML5, meeting client specifications.

STRENGTHS

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|------------------|--------|-----------------------|--------------|
| • Linux | • Rust | • Distributed Systems | • Kubernetes |
| • Authentication | | | |

INDUSTRY EXPERTISE

Backend

Distributed Systems

Full-Stack

Cybersecurity

TECHNICAL SKILLS

Programming Languages

Rust, Python, Typescript, NodeJs, C, Java

AWS

SQS, Lambda, Aurora RDS, DynamoDB, S3, Redshift, IAM, DataDog, CloudWatch, CloudFront, VPC, EC2, Secrets Manager, KMS, SNS, EventBridge, EKS

Backend

Docker, Kubernetes, Event Driven Architecture, Web Sockets, Distributed Systems, REST APIs, TDD, Microservice Architecture, Asynchronous Programming, Parallel Programming, gRPC

Cryptography

Public Key Cryptography, Symmetric Encryptions, RSA, PGP, AES, ECDH, Homomorphic Encryption, FHE, TFHE, ZK-SNARKs, ZK-STARKs

Frameworks

Tokio, Rocket-rs, Axum, Actix, Node.js

Frontend

React, Yew-rs, HTML, CSS, Javascript, Tailwind, Vite, Webpack, WebAssembly

Developer Tools

Git, Gitlab, Github, Trunk, Cargo, Jira, Confluence

CICD

Github Actions, Gitlab CICD, Terraform, Docker, Kubernetes

Databases

PostgreSQL, MySQL, Relational Database Design, MongoDB, Redis

Messaging and Distributed Systems

Kafka, Redis Streams

Observability

Prometheus, Grafana, Metrics, Logging, Tracing

Others

TDD, Agile Methodology, Linux

Education

Faculdade Anhanguera, Divinópolis

Bachelor's in Computer Science, Jan 2018 to Dec 2022

- Learned the foundations of computer science.
- Coded in Java, Python and the standard web stack.
- Studied Databases, Systems architecture, best practices for system's development, cybersecurity, hardware, etc.
- My final paper was a study on the advantages of using the Rust programming language for system's development.

SENAI Anielo Greco, Divinópolis, Minas Gerais, Brazil

Associate's Degree in Industrial Electronics Technology/Technician, Jun 2013 to Dec 2014

- Practice building industrial electrics and electronics.
- Set up electrical systems.
- Built electronic circuits with compiled C code for automation, etc.
- Implemented Arduino, Pic and Raspberry development.
- Configured Electric motors and Industrial Electric controllers for automation.

Links

[Linkedin](#), [Website](#), [Github](#)

Languages

English, Portuguese